Mathematical supplement to Translation confinement and angular propagation for a Reuleaux test family in Lebesgue's covering problem

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Abstract

This supplement records the scope and support mechanism of inherited restricted-cell and endpoint certificates, together with exact data for the conditional angular application in the main note. A computational companion supplies finite checks of the recorded witnesses. Restricted vertical domains remain restricted; the endpoint pilot is numerically subsumed by the known all-placement 0.8344 bound.

S1 Notation and inherited evidence

Write $\mathbb{D}(p,r) = \{x \in \mathbb{R}^2 : \|x-p\| \le r\}$ and let $D = \mathbb{D}(0,1/2)$. For $n = 3, 5$, put

$$\rho_n = \frac{1}{2\cos(\pi/(2n))}, \qquad c_{n,j} = \rho_n(\cos(2\pi j/n), \sin(2\pi j/n)), \qquad B_n = \bigcap_{j=0}^{n-1} \mathbb{D}(c_{n,j}, 1).$$

For independent translations $t_n = (x_n, y_n)$ and rotation R_ϕ,

$$H(t_3, t_5, \phi) = \operatorname{area}\operatorname{conv}\big(D \cup (t_3 + B_3) \cup (t_5 + R_\phi B_5)\big), \qquad G(\phi) = \inf_{t_3, t_5 \in \mathbb{R}^2} H(t_3, t_5, \phi).$$

The main note proves confinement at $L_3 = 35454253/42400000$, with radii $d_3 = 4/25$ and $d_5 = 9/100$, and a Lipschitz bound $C = 4270068/1750000$ for $\min\{L_3, G\}$. Its target threshold is $\tau = 1669/2000 = 0.8345$. The relevant vertical rectangle is

$$V = [-4/25, 4/25] \times [-9/100, 9/100].$$

A regional bound $G \ge L$ on a nonempty closed interval I implies

$$G(\phi) \ge \min\{L, L_3\} - C \operatorname{dist}(\phi, I). \tag{S1}$$

The word "regional" in this implication restricts the angle only: the premise quantifies over every translation pair.

The numerical witnesses below are inherited records. The computational companion supplies their mathematical data and checks point admissions, selected cycles and polynomial inequalities on the stated domains. The arguments below explain how those finite checks support the bounds, including the assumptions and quantifier restrictions that remain.

S2 Restricted certificates and their support mechanism

The following standard convexity mechanism explains why the retained certificates eliminate both horizontal translations, while leaving their vertical quantifiers restricted.

Lemma S2.1 (Common-direction convexity)**.** *Let P_0, P_3, P_5 be fixed finite planar point sets. Fix the vertical translations of P_3, P_5. The area of their hull with P_0 is jointly convex in the two horizontal translations. The analogous assertion holds for joint vertical translations when the horizontal translations are fixed.*

Proof. Restrict the two horizontal parameters to an arbitrary affine line $(u, v) = (u_0, v_0) + s(\alpha, \beta)$. Points of P_0 are stationary; points of P_3 and P_5 move with constant speeds α and β in the common horizontal direction. Their hull is a shadow system. One may choose the parameter origin away from the finitely many crossings of distinct trajectories and merge identical trajectories; thus coincident point labels do not affect this description. Area here means ambient two-dimensional measure, including degenerate hulls. The classical convexity theorem for shadow-system volume [8, Section 3, Theorem 6], also recalled in [7, Introduction, p. 2] therefore gives a convex area function of s. Convexity on every parameter line is joint convexity in (u, v). Interchanging the coordinate axes proves the vertical assertion. □

An admitted finite point set lies in its actual body for every angle of its closed source interval. At a chosen horizontal anchor a selected cycle is checked strictly convex. If the cycle persists locally, its shoelace area is affine in the two horizontal translations: potential quadratic terms vanish because every motion is in that same direction. This is a global supporting plane for the selected hull's convex area function. To justify this step directly, a convex function locally equal to an affine function ℓ satisfies $f(z) \geq \ell(z)$ everywhere: apply convexity along a short segment from the anchor towards z and use the local equality there. Inclusion makes that plane a lower bound for the full actual hull even if unselected points become extreme.

In the restricted-cell constructions, exact orientation determinants are affine in (y_3, y_5) at a fixed horizontal anchor. Strict corner tests therefore preserve the selected cycle on an entire closed rectangle. Supporting planes have coefficient form

$$c_i = c_{i0} + d_{i3}y_3 + d_{i5}y_5, \qquad a_i = a_{i0} + k_i y_5, \qquad b_i = b_{i0} - k_i y_3,$$

where a_i, b_i multiply x_3, x_5. For a triple of planes define cyclic $N_i = a_j b_k - a_k b_j$ and $\Delta = \sum_i N_i$. The bilinear terms cancel, so N_i, Δ are affine in the vertical pair. Exact identities

$$\sum_i N_i = \Delta, \qquad \sum_i N_i a_i = 0, \qquad \sum_i N_i b_i = 0$$

and corner tests $\Delta > 0$, $N_i \geq 0$ supply nonnegative weights N_i/Δ canceling both horizontal slopes on the whole cell. The polynomial $T = \sum_i N_i c_i - \tau\Delta$ is checked in its tensor-product Bernstein basis on the rectangle. Since the basis functions are nonnegative and sum to one, positive coefficients imply $T > 0$ there. A positive coefficient minimum divided by a corner maximum of Δ supplies an exact strict margin. Numerical linear programming supplies proposals; only the exact admissions and identities establish these bounds.

The retained A statement is

$$H((x_3, y_3), (x_5, y_5), \phi) \geq L_A > \tau \quad \text{for } \phi \in [49/10000, 37/5000], \quad x_3, x_5 \in \mathbb{R}, \quad (y_3, y_5) \in \mathcal{A}, \quad \text{(S2)}$$

where $\mathcal{A}$ is the union of exactly 33 closed rectangles in the companion's restricted-cell data; its exact L_A is given in Section S3. The 33 certified cells and 24 residual closed leaves cover V with

shared faces. Sixteen residual leaves are central and eight are outer. The residual leaves form an obligation cover: shared faces already belonging to certified neighbors are not thereby declared uncertified. This union does not exhaust V by certified cells, so (S2) supplies neither a G bound nor the hypothesis of (S1).

The retained B construction gives two further statements of exactly the same unrestricted-horizontal, restricted-vertical form. Their closed angle intervals and closed vertical rectangles W_1, W_2 are

cell	closed angle interval	closed vertical rectangle W_i
$i = 1$	$[\frac{101}{10000}, \frac{111}{10000}]$	$[\frac{-2049}{1000000}, \frac{-1849}{1000000}] \times [\frac{-353}{500000}, \frac{-253}{500000}]$
$i = 2$	$[\frac{99}{1000}, \frac{101}{1000}]$	$[\frac{-9733}{1000000}, \frac{-9533}{1000000}] \times [\frac{-9191}{1000000}, \frac{-8991}{1000000}]$

For each row i, every angle in its interval, every vertical pair in W_i, and every independent $x_3, x_5 \in \mathbb{R}$ satisfy $H \geq L_{B,i} > \tau$. The exact constants are in Section S3. Neither row supplies a true-G premise.

A separate B endpoint pilot yields the weaker all-translation bound

$$H(t_3, t_5, \phi) \geq L_B < \tau \quad (t_3, t_5 \in \mathbb{R}^2), \qquad \phi \in [\pi/15 - 9/1000, \pi/15 - 1/125], \tag{S3}$$

with L_B approximately 0.8341743532983762. Mishra's all-placement bound 0.8344 for the same family already implies this numerical conclusion [2]. We retain the pilot only as a method illustration. Here the elimination of vertical translations uses only a paired, admitted inner family. Indeed a common rotation by $-2\pi/3$ fixes D, B_3 and, using the $2\pi/5$ pentagon period, sends the relative pentagon phase to $\phi + 2\pi/15 = \pi/5 + z$, where $z = \phi - \pi/15$. Since $\pi/5$ and π differ by $4\pi/5$, $R_{\pi/5+z} B_5 = R_z(-B_5)$. The pilot has $z \in [-9/1000, -1/125]$.

Its finite polygons P_D, P_3, P_5 have exact horizontal reflection closure, and are admitted respectively inside D, B_3 and $\bigcap_{|z| \leq 9/1000} R_z(-B_5)$. Let $f(u, y_3, v, y_5)$ be their hull area. For fixed u, v, Lemma S2.1 gives convexity in the vertical pair. Reflection closure of these inner polygons gives $f(u, y_3, v, y_5) = f(u, -y_3, v, -y_5)$, so Jensen's inequality yields

$$f(u, y_3, v, y_5) \geq f(u, 0, v, 0).$$

Exact nonnegative combinations of horizontal supporting planes at zero vertical translations then cancel both horizontal slopes, giving $f(u, 0, v, 0) \geq L_B$. Inclusion and the common rotation prove the full translation quantifiers in (S3). This argument asserts reflection symmetry only for the admitted inner polygons. It does not assert symmetry of actual optimizers or convexity in all four translation coordinates jointly. The deficit $\tau - L_B > 0$ is a certificate deficit, not an upper bound on the actual infimum. It adds no threshold-τ coverage in this application.

Continuous rotating-disk admission in the inherited witnesses includes possible interior extrema. For a point p and a defining center c, the dot product during rotation is $A \cos s + B \sin s$; besides endpoint values its possible interior minimum is $-\sqrt{A^2 + B^2}$. An uncertain stationary candidate is retained, rather than omitted. This admission condition is part of the saved geometry evidence; the preceding support arguments alone do not recreate it.

S3 Exact values and closed cell domains

For the three all-translation premises in the main note, the closed angle intervals are $I_0 = [0, 3/1250]$, $I_{10} = [99/10000, 101/10000]$ and $I_{\text{end}} = [\pi/15 - 1/125, \pi/15]$. Their constants satisfy $\tau < L_i < L_3$,

and $\eta_i = (L_i - \tau)/C$. All rational constants below are specified by exact numerator and denominator strings in the companion's mathematical data. Exact values remain authoritative over decimal displays. The following displays factor powers of ten in the denominators and use explicit integer sums for the two longest numerators. Each decomposition represents the corresponding exact rational value.

$$L_0 = \frac{\mathcal{N}_0}{15861207047746954056868474314175890070101275 \cdot 10^{46}}, \qquad \text{where}$$

$$\begin{aligned}\mathcal{N}_0 = {} & 13236196765582091232562705 5417 \cdot 10^{60} \\ & + 87395527889703921181723329 0204 \cdot 10^{30} \\ & + 13945865466551668496320896 2537\end{aligned}$$

$$L_{10} = \frac{795349321566328674778388273400918884961583736 3}{9530786886710063553288 \cdot 10^{24}}.$$

$$L_{\mathrm{end}} = \frac{\mathcal{N}_{\mathrm{end}}}{12896863427523737699213753454177288325305459210 5 \cdot 10^{47}}, \qquad \text{where}$$

$$\begin{aligned}\mathcal{N}_{\mathrm{end}} = {} & 10763 \cdot 10^{90} \\ & + 67406538188192623279086400 7209 \cdot 10^{60} \\ & + 75407972074008911280926990 2630 \cdot 10^{30} \\ & + 38588899475126330555915722 0541\end{aligned}$$

$$L_A = \frac{706650915209303190752170802264824942916682856 7}{8467935872589912289148 \cdot 10^{24}}.$$

$$L_{B,1} = \frac{247075393635668115564763143381461722424251 47}{295996811788273683874 \cdot 10^{23}}.$$

$$L_{B,2} = \frac{227499933714523570019636401400476784809266499 7}{2724179336032307676194 \cdot 10^{24}}.$$

$$L_B = \frac{210803804668486022720529125627485392569560569}{252709525095029507606875 \cdot 10^{21}}.$$

The exact halo radii and the positive endpoint deficit are specified by

$$\eta_i = \frac{L_i - \tau}{C}, \qquad i = 0, 10, \mathrm{end}, \qquad \tau - L_B > 0.$$

The companion retains the exact bound and halo-radius fractions. The exact added and total covered lengths in the main note have the readable symbolic representations

$$\ell_{\mathrm{added}} = \eta_0 + 2\eta_{10} + \eta_{\mathrm{end}}, \qquad \ell_{\mathrm{covered}} = \frac{53}{5000} + \ell_{\mathrm{added}}.$$

Their expanded numerator and denominator strings are retained in the companion's propagation data. The exact remaining total length is $\pi/15 - \ell_{\mathrm{covered}}$. The data also contain affine-π gap endpoints and widths and rational radius brackets. The checker reports outward 256-bit Arb displays of the gap widths. The 33 certified vertical cells are listed below. Their shared faces are included; their union is not all of V. The certificate data also retain the 24 residual closed leaves and each cell's individual bound and selected witness. All cells in this table use the common closed angle interval $[49/10000, 37/5000]$.

cell	y_3 interval	y_5 interval
1	$[\frac{-7}{512}, 0]$	$[\frac{21}{2560}, \frac{63}{5120}]$
2	$[0, \frac{7}{512}]$	$[\frac{21}{5120}, \frac{21}{2560}]$
3	$[0, \frac{7}{512}]$	$[\frac{21}{2560}, \frac{63}{5120}]$
4	$[\frac{-7}{512}, \frac{-7}{1024}]$	$[0, \frac{21}{5120}]$
5	$[\frac{-7}{512}, \frac{-7}{1024}]$	$[\frac{21}{5120}, \frac{21}{2560}]$
6	$[\frac{7}{1024}, \frac{7}{512}]$	$[0, \frac{21}{5120}]$
7	$[\frac{-7}{1024}, \frac{-7}{2048}]$	$[\frac{21}{5120}, \frac{21}{2560}]$
8	$[\frac{-7}{1024}, \frac{-7}{2048}]$	$[\frac{21}{10240}, \frac{21}{5120}]$
9	$[\frac{-7}{2048}, 0]$	$[\frac{21}{5120}, \frac{63}{10240}]$
10	$[\frac{-7}{2048}, 0]$	$[\frac{63}{10240}, \frac{21}{2560}]$
11	$[\frac{21}{2048}, \frac{7}{512}]$	$[\frac{-21}{5120}, \frac{-21}{10240}]$
12	$[\frac{7}{2048}, \frac{7}{1024}]$	$[\frac{21}{10240}, \frac{21}{5120}]$
13	$[\frac{-21}{2048}, \frac{-7}{1024}]$	$[\frac{-21}{5120}, \frac{-21}{10240}]$
14	$[\frac{-21}{2048}, \frac{-7}{1024}]$	$[\frac{-21}{10240}, 0]$
15	$[\frac{-7}{1024}, \frac{-7}{2048}]$	$[\frac{-21}{5120}, \frac{-21}{10240}]$
16	$[\frac{-7}{1024}, \frac{-7}{2048}]$	$[\frac{-21}{10240}, 0]$
17	$[\frac{-7}{2048}, 0]$	$[\frac{21}{10240}, \frac{21}{5120}]$
18	$[\frac{7}{2048}, \frac{7}{1024}]$	$[\frac{-21}{5120}, \frac{-21}{10240}]$
19	$[\frac{7}{2048}, \frac{7}{1024}]$	$[\frac{-21}{10240}, 0]$
20	$[\frac{7}{1024}, \frac{21}{2048}]$	$[\frac{-21}{5120}, \frac{-21}{10240}]$
21	$[\frac{7}{1024}, \frac{21}{2048}]$	$[\frac{-21}{10240}, 0]$
22	$[\frac{21}{2048}, \frac{7}{512}]$	$[\frac{-21}{10240}, 0]$
23	$[0, \frac{7}{2048}]$	$[\frac{21}{10240}, \frac{21}{5120}]$
24	$[\frac{7}{2048}, \frac{7}{1024}]$	$[0, \frac{21}{10240}]$
25	$[\frac{-7}{512}, \frac{-49}{4096}]$	$[\frac{-21}{5120}, \frac{-21}{10240}]$
26	$[\frac{-49}{4096}, \frac{-21}{2048}]$	$[\frac{-21}{5120}, \frac{-21}{10240}]$
27	$[\frac{-7}{512}, \frac{-49}{4096}]$	$[\frac{-21}{10240}, 0]$
28	$[\frac{-49}{4096}, \frac{-21}{2048}]$	$[\frac{-21}{10240}, 0]$

cell	y_3 interval	y_5 interval
29	$[\frac{-7}{2048}, \frac{-7}{4096}]$	$[\frac{-21}{5120}, \frac{-21}{10240}]$
30	$[\frac{7}{4096}, \frac{7}{2048}]$	$[0, \frac{21}{10240}]$
31	$[\frac{-7}{4096}, 0]$	$[\frac{-21}{5120}, \frac{-63}{20480}]$
32	$[\frac{-7}{1024}, \frac{-21}{4096}]$	$[\frac{21}{20480}, \frac{21}{10240}]$
33	$[\frac{-21}{4096}, \frac{-7}{2048}]$	$[\frac{21}{20480}, \frac{21}{10240}]$

Computational companion and scope. The finite checking programs and their required witness data are available in the Universal covering computational companion (version `companions-v2`). Its README gives the run commands, arithmetic dependencies and coverage of each check. Exact integer and rational arithmetic is combined with outward Arb/FLINT enclosures at 256 bits for the numerical geometry. The programs verify recorded witnesses and compute the propagation constants and remaining gaps. The regional implications also use the analytic confinement argument in the main note and the support arguments above. The main note retains its conditional formulation of the three all-translation premises. The 33 A cells and two B cells keep their restricted domains, the endpoint pilot remains below τ, and the two angular gaps remain open.

References

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